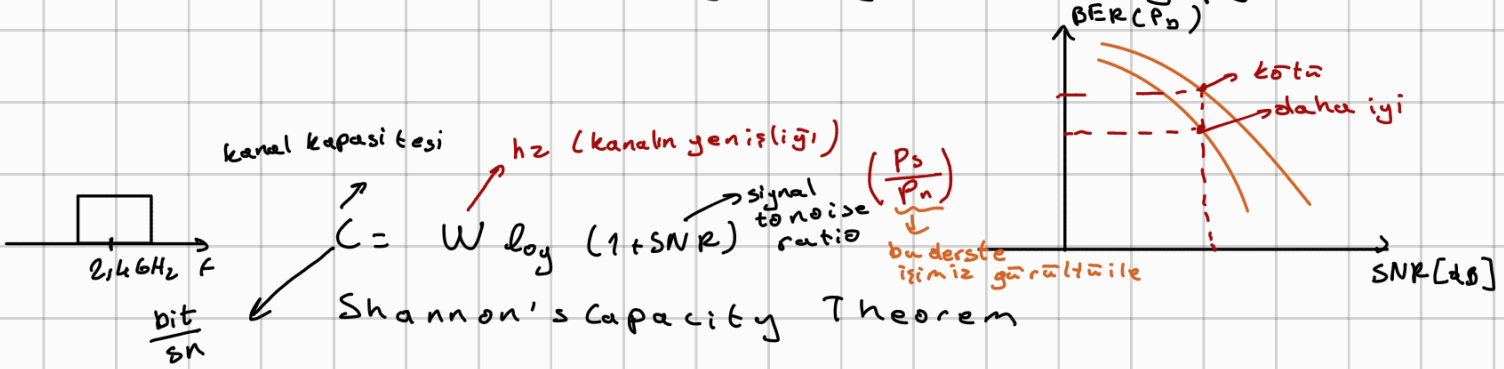
Selami Şahin
0536 563 8298
selsah80@gmail.com

1 - RF Basic Concepts & components
2 - RF design Theory & principle
3 - RF design Architecture (Rx & Tx)
4 - Modulation in Comm. Sys.
5 - RF Microwave & Transmissions line
6 - RF & Microwave Filters
7 - RF Networks (S parametreleri, saçılma ...)
8 - RF components (Mixers, switches, Oscilator, ...)

} 4 özellikle en iyi anlamamız gerekiyor

%30 Arasınav
%10 ödev (4-5 adet)
%40 final
%20 Project → Bir tane makale seçiyorsunuz, ve bu makaleyi sunuyorsunuz, sunum yapıyorsunuz.

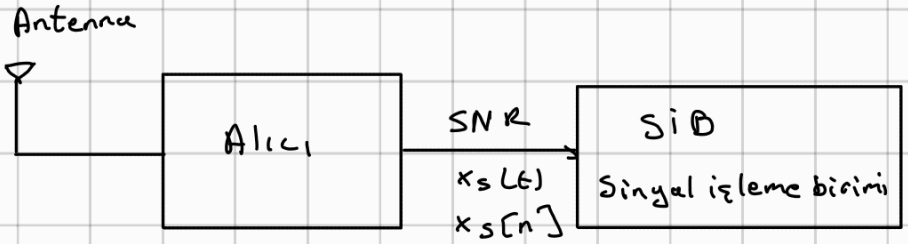




Televizyon 10GHz

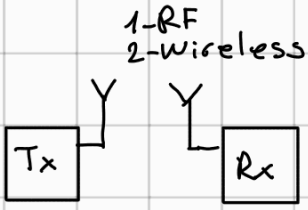
ITU kurumu nerde haberleşeceğini söyler

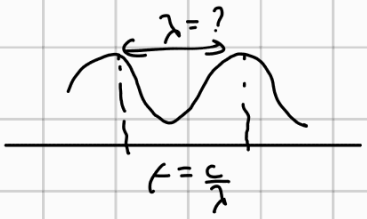
RF Işıma yapar, anten çalıştığı bölgeye göre tasarlanmış olacak.



Gürültüye minimalize etmeliyiz

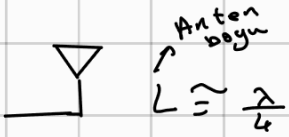
RF: 3khz frekansından 300 GHze kadar olan tüm Sinyaller.
(100 GHz üstü milimitre wave, 10GHz üstü microwave)





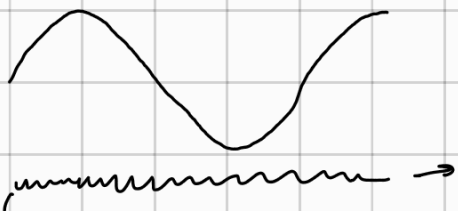
	f	$\lambda = \frac{c}{f}$	
VLF: (Very low frequency)	3 - 30khz	33 km ~ 10 km	Maritime radio navigation (Denizcilikte çok uzun mesafe haberleşme)
LF:	30 ~ 300k	10 km ~ 1km	
MF:	300k ~ 3M	1km ~ 100m	AM Radio aviation navigation
HF:	3M ~ 30M	100m ~ 10m	
VHF:	30M ~ 300M	10m ~ 1m	→ TV, FM
UHF:	300M ~ 3GHz	1m ~ 100 mm	→ TV, mobile, GPS, Wi Fi, 4G
SHF:	3G ~ 30GHz	100 mm ~ 10mm	→ Wifi, satatellite
EHF:	30G ~ 300GHz	10mm ~ 1mm	→ Astronomy satelite

radio



frekans büyüdükçe anten küçülür.

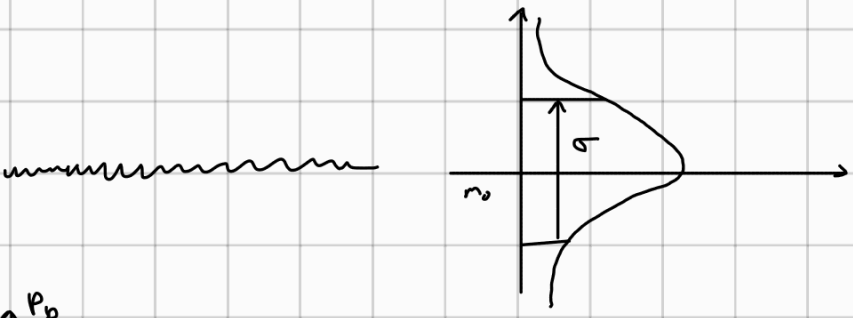
Noise:

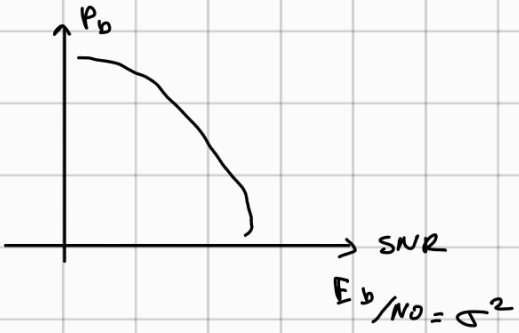


→ Rastgele süreç

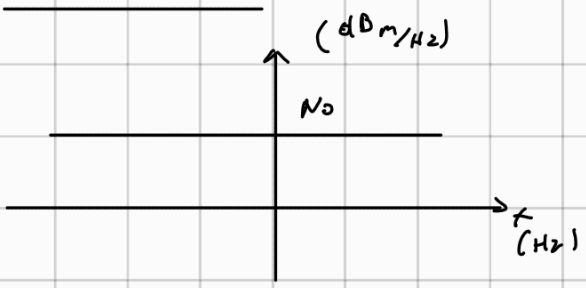
↳

Vargan gücü

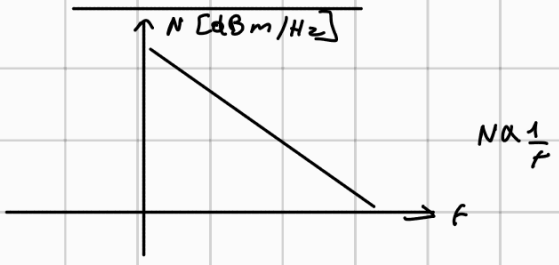




White Noise



Pink noise

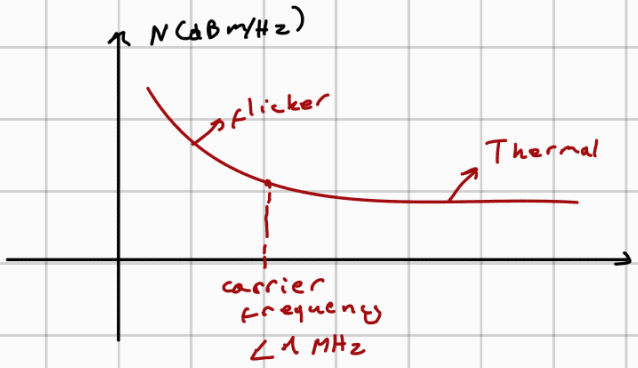


White Noise ~ Thermal Noise (Shot Noise) [→ White Noise N(mo↑)]

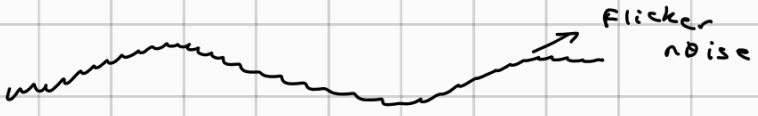
Flicker Noise ($\frac{1}{f^n}$ noise) ~ Pink noise

(Low frequency)

+ üst üste çizersek



SNR oranını



$$SNR = \frac{P_s[W]}{P_n[W]} \quad [\,] \text{ birimsiz}$$

$$SNR_{[dB]} = 10 \log\left[\frac{P_s}{P_n}\right] \; dB$$

sadece logaritma birimi bel
10 ile çarparsanız decibel

$$\log[SNR]$$

$$10 \log[2 \cdot SNR] = \underbrace{10 \log 2}_{3dB} + 10 \log[SNR]$$

2 kat arttırırsak 3 db

10 kat 10 db ✯S

100 kat 20 db

10 W → dBW

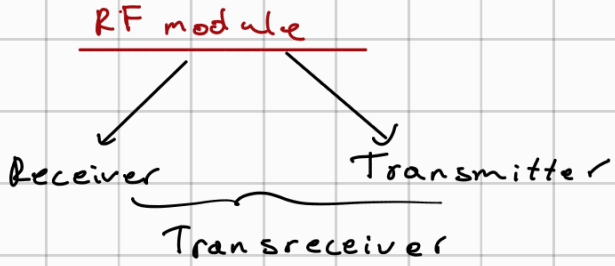
$10 \log\left[\frac{P_s}{1W}\right] \Rightarrow$ dBW

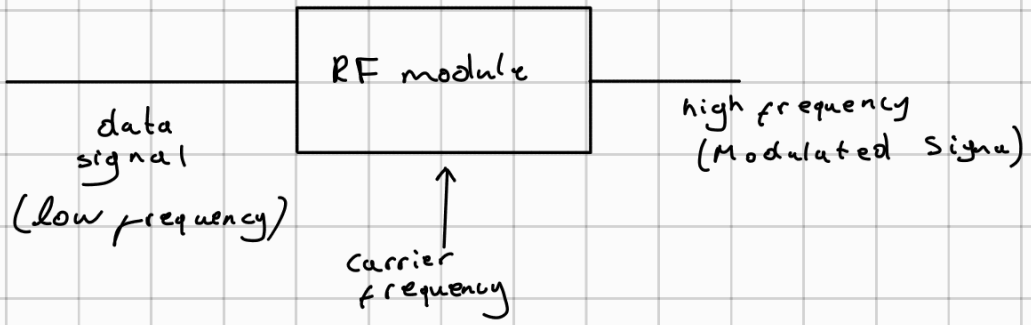
$10 \log\left[\frac{P_s}{1mW}\right] \Rightarrow$ dBm

$$10 \log[P_s] + 10 \log\left[\frac{1}{1mW}\right] = \left(P_s[dBw] + 30\,dB\right)[dB_m]$$

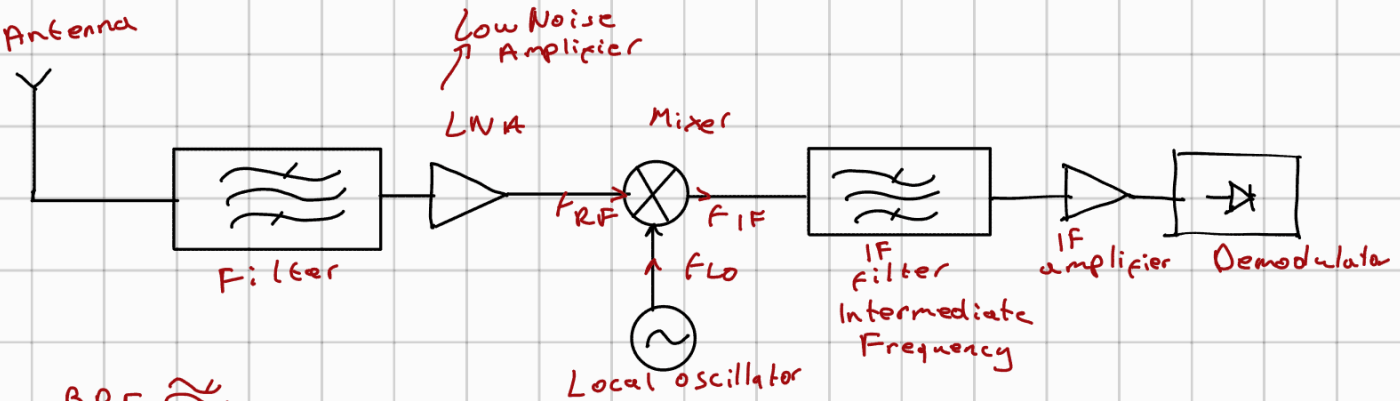
$$10 \log\left[\frac{1mW}{1mW}\right] = 0 dBm$$

$$10 \log\left[\frac{10mW}{1mW}\right] = 10 dBm$$





Reciever



BPF

LPF

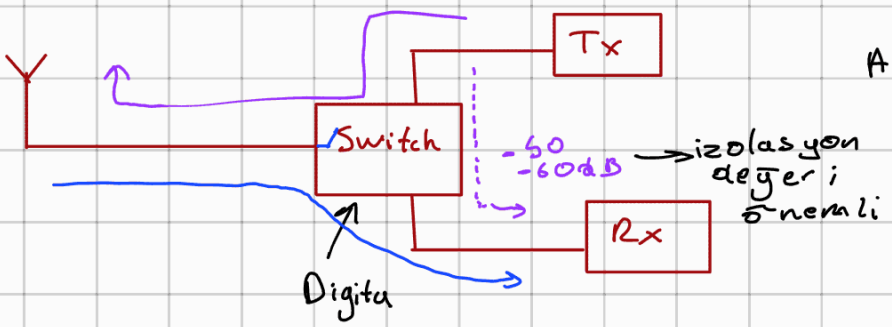
HPF

BSP

RF Front-End

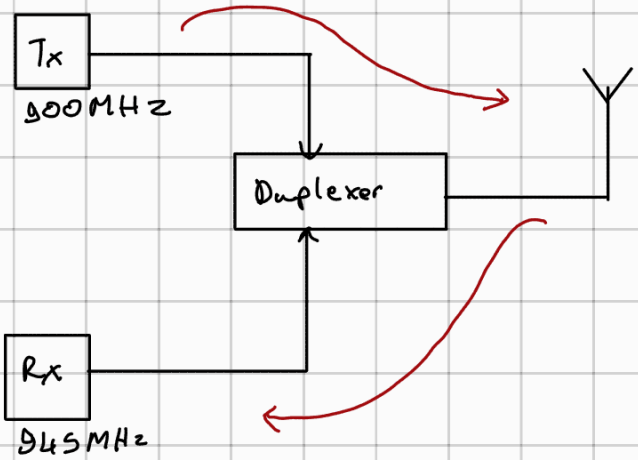
||

RFE

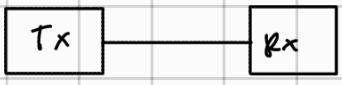


Aynı anda hem verici hem alıcı çalışmaz

Half duplex



Full duplex

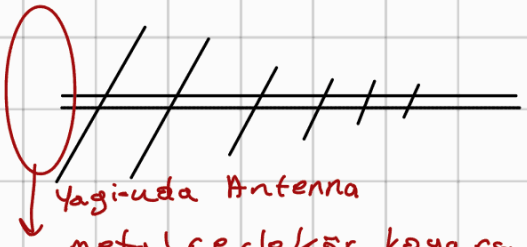


Simplex

Antennas

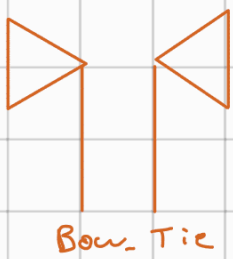
Log - periodic Antenna

heriki yerdon alır

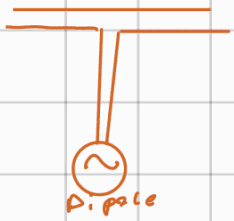


Yagi-uda Antenna

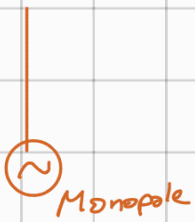
metal reflektör koyarsanız sadece tek yönlü ← olur



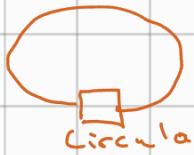
Bow Tie



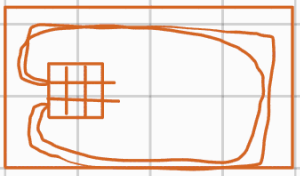
Dipole



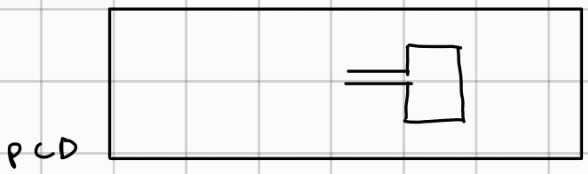
Monopole



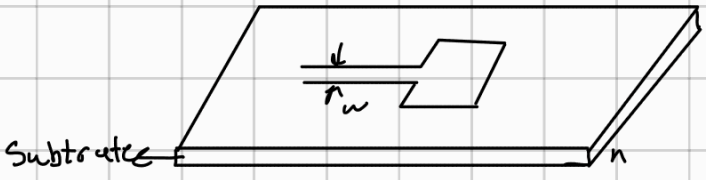
Circular



Patch Antenna



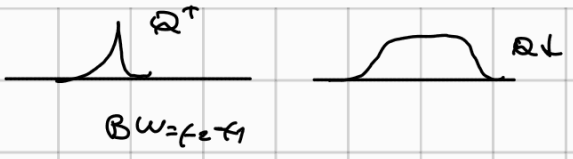
PCD



Substrate

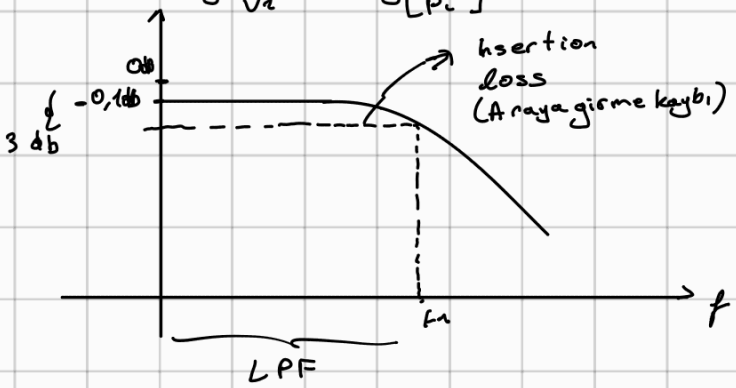
n

RF Filters

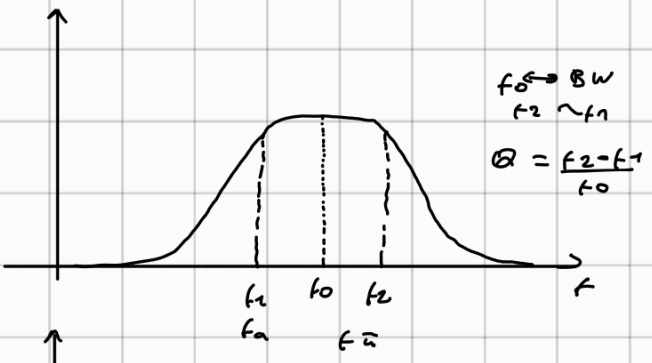


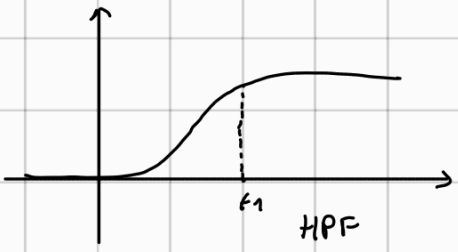
$BW = f_2 - f_1$

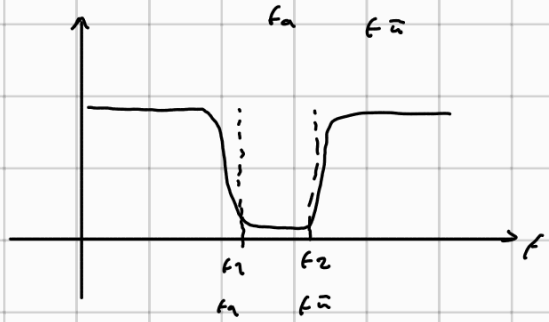
$20\log\frac{V_o}{V_i} = 10\log\left[\frac{P_o}{P_i}\right] = 6$



LPF





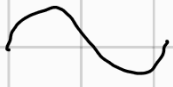


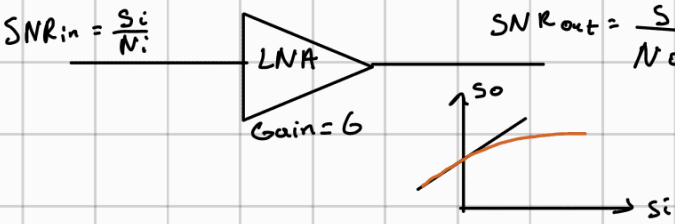
BSF

Aktif komponentler akımı yükseltiyor.

hep bir dirence pasif yani

LNA





$N_A \lll$ seçilmeli

Filtrenin kendisinden kaynaklanan ek gürültü

$SNR_{out} < SNR_{in}$

Requirements

- Low noise coming from device ($\Rightarrow$ low NF) → Noise Figure
- Linearity ($\Rightarrow$ high P_1dB → 1dB Compression Point; IP3 → Third order intercept Point)
- Low Power Consumption
- Low Area " (Small form factor)
- Gain ($\Rightarrow S_{21}$)
- Matching to anten filter